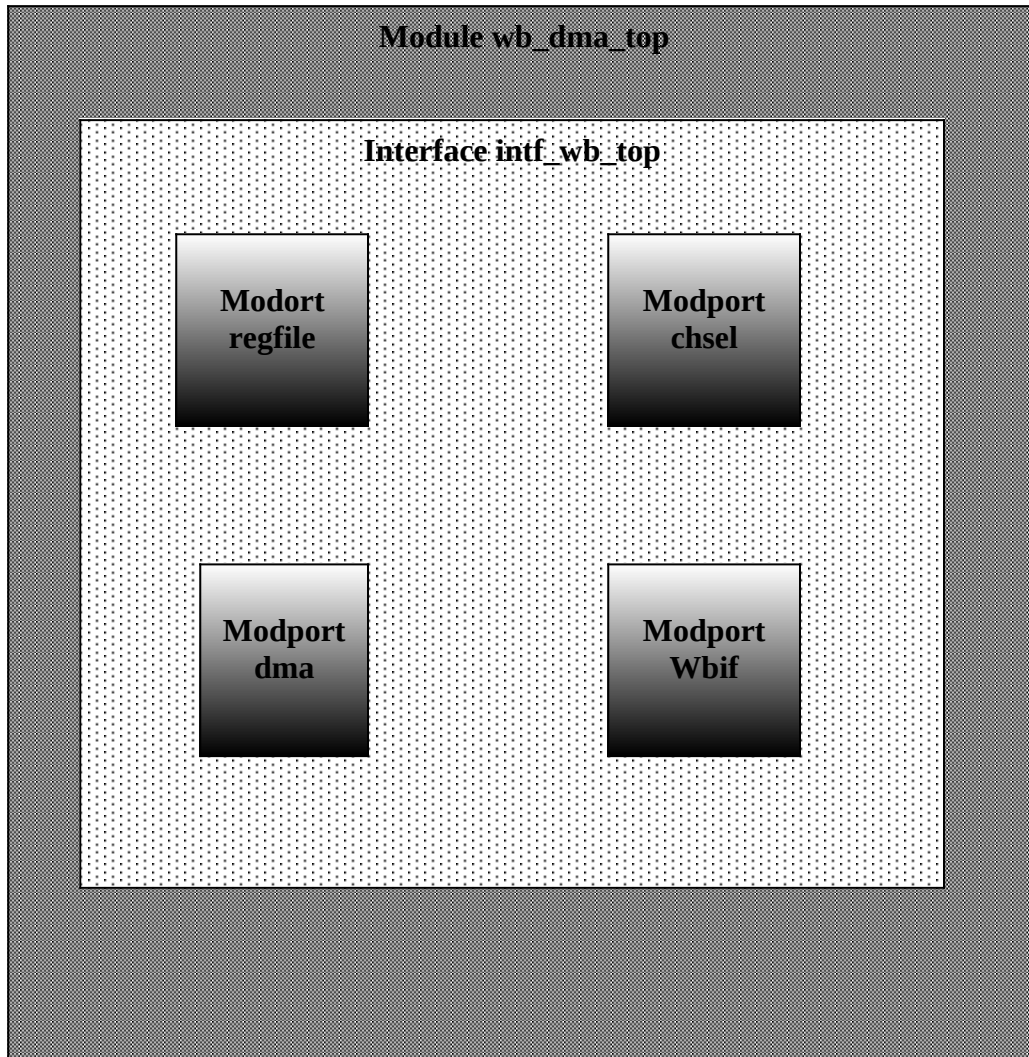


Wishbone dma design document

The wishbone dma is designed using system verilog 3.0. The purpose of design is to show the design characteristics of system verilog and to demonstrate the reduction of line count using system verilog.

The top-level wb_dma_top has instance of interface intf_wb_dma_top. The interface has regfile, ch_select, dma and wishbone interface as modports and also in out ports from to interface. All the modports are instantiated and interconnected.



The design demonstrates the use of several features of system verilog like, structures, interface, modports, logic, bit, multi dimensional arrays, ++, --, always_ff, always_comb, type defs, enumeration types, generate, unique in case, priority in ifelse, fork join blocks etc.

Name of module	Features used
Wb_dma_top	Instantiation of interface and its modports
Intf_dma_top	Interface, modport, type_defs, structure
Dma	Generate, always_ff, always_comb, always_latch, enum
Wb	Interface, modports, generate, always_comb, always_ff
Channel select	Generate,always_comb,always_ff, unique, priority
Regfiles	Interface, modports, generate, always_comb, always_ff, Usage of structures, multi-dimensional arrays

Name of module	Verilog	System verilog
Wb_dma_top	1000	182
Dma	400	298
Wb	1000	88
Channel select	3200	397
Regfiles	1000	400
Total	6600	1365

The directory structure is arranged as follows in cvs

```
wb_dma_core/rtl/verilog    -- code from open cores
wb_dma_core/rtl/sysverilog -- design directory
rtl/sysverilog/top         Top modules
rtl/sysverilog/chsel       Channel select modules
rtl/sysverilog/dma         DMA modules
rtl/sysverilog/regfile     Register modules
rtl/sysverilog/wb          Wishbone modules
```

documentation is in

```
doc/top                    Top modules
doc/chsel                  Channel select modules
doc/dma                    DMA modules
doc/regfile                register modules
doc/wb                     Wishbone modules
```

sim directory **From where the simulations are executed.**

In this directory, to run sysverilog, just use wbrun script.

To run verilog code goto sim/verilog use make clean and make.

Test Bench --- Existing test bench from opencores.